Lecture 7: Conscious of the Present; Conscious of the Past: Language (cont.); Vision and Memory

Introduction to Psychology · Yale University, Spring 2007 · Professor Paul Bloom

Study guide based on the lecture transcript. Source: https://oyc.yale.edu/psychology/psyc-110/lecture-7 (CC BY-NC-SA 3.0).

Before you start

This lecture first finishes the topic of language: do animals have language, and how is language related to thought? Then Professor Bloom starts a new topic: perception, attention and memory. His main message is that we are often wrong about our own experiences.

Learning goals

After this lecture you should be able to:

- Describe three kinds of animal communication and how they differ from human language.
- Explain why seeing is hard and how the brain uses unconscious assumptions.
- Give examples of depth cues and explain one visual illusion.
- Describe the main types of memory.
- Explain what the Stroop effect and change blindness show about attention.

Key ideas

1. Non-human Communication

- Follow-ups from Lecture 6: newborns already prefer their own language, so they probably hear its rhythms in the womb. The man in the "Let him have it" case was found guilty, executed in 1957 and pardoned in 1988.
- Animals clearly have communication systems, but almost certainly not language in the narrow sense of Lecture 6.
- There are three kinds: a finite list of calls (vervet monkeys warn about a snake or a leopard); a continuous signal (in the bee dance, a more intense dance means a richer food source); and random variation on a theme (birdsong). None has phonology, morphology, syntax or arbitrary names.
- Trained primates such as Kanzi and Nim Chimpsky learn few words after years of training. Their word order is limited, they lack recursion, and they repeat a lot: "Me banana, you banana, me you give."
- Two weak arguments: "chimps are smart" (being smart is not enough for language) and "chimps are our nearest relatives" (we split about five million years ago, enough time for language to evolve). Bloom's personal view: studying animal communication in the wild is more interesting than teaching apes a human language.

2. The Relationship between Language and Thought

- Is language necessary for abstract thought? We can test this by studying babies and chimpanzees, who have no language.
- Does your language affect how you think? This claim is called **linguistic relativity** or the **Sapir-Whorf hypothesis**. Researchers compare speakers of languages like English and Korean.
- Bloom also notes that reading, unlike speech, is a cultural invention that is learned with great difficulty.

3. Question and Answer on Language

Nobody really knows why some people learn languages faster; girls are slightly ahead, and genes play a big part. Children who hear two languages learn both, reach milestones on time, and show no known harm. In right-handed people, language is probably in the left side of the brain; in left-handed people it varies.

4. Introduction to the Complexity of Perception and Expectation

- Bloom's claims: perception uses educated, unconscious guesses; we miss a lot of what happens; and you can't trust some of your memories. Psychologists asked people on September 12, 2001, where they had heard the news. A year or two later, their stories were often wrong.
- Seeing is very hard. In 2007, Bloom said no machine could recognize people and objects as well as a one-year-old. Imagining a "little man watching a TV in your head" explains nothing; it only pushes the question back.
- The retina is a 2-D surface of nerve cells that fire or don't fire. Many 3-D scenes can make the same 2-D image, so the brain uses unconscious assumptions to work out the 3-D world.
- **Brightness:** we assume shadows make surfaces darker, so we see a surface in shadow as lighter. Two blocks look different but are the same; cover the background and they look the same.
- **Objects:** Gestalt principles are cues that group parts into objects, for example proximity (things close together belong together), similarity and common movement (things that move together are one object). They can fool us: we see a triangle that is not there.
- **Depth:** cues include binocular disparity (each eye gets a slightly different image), interposition (a nearer thing blocks part of a farther thing), relative size, texture gradient (texture looks smaller far away) and linear perspective (parallel lines seem to come together in the distance).
- In the Mueller-Lyer and Ponzo illusions, depth cues make one line seem farther away, so the brain sees it as longer. Roger Shepard's two tables look different but are the same size.

5. Linking Attention and Memory

- Memory is broad: knowing English and knowing how to chew are memories too. Amnesia can remove memories of the past, or the ability to form new ones. A film showed a choir director with the second type, after brain damage to his temporal lobes and hippocampus.
- Types: **sensory memory** (an afterimage; echoic memory keeps a few seconds of sound), **short-term or working memory** (a few minutes), and **long-term memory**. Also **explicit** (conscious) vs. **implicit** (unconscious), and **semantic** (facts) vs. **episodic** (your own life events). Memory has three steps: encoding (getting it in), storage (holding it) and retrieval (getting it out).

- **Attention** decides what gets into memory, like a spotlight. A green "x" among black ones is found at once; a red "o" among black o's and red x's takes effort.
- The **Stroop effect:** naming the ink color of color words printed in a different color is hard, because skilled readers cannot stop reading.
- In the gorilla video, about 50% of first-time viewers who count basketball passes miss a person in a gorilla suit. In **change blindness** studies by Dan Simons, most people did not notice when the stranger asking them for directions was replaced by another man. We miss these changes, but we think we see the world as it is.

6. Question and Answer on Attention and Memory

When a skill like reading is highly practiced, it becomes automatic and you cannot stop it; Bloom does not know why. The choir director still remembered many things from before his illness, such as his name.

Key vocabulary

Term	Meaning in simple English	Example from the lecture
linguistic relativity	the idea that your language affects how you think (Sapir-Whorf hypothesis)	studies of English and Korean speakers
retina	the layer of nerve cells at the back of the eye	a 2-D surface; the brain must infer a 3-D world
Gestalt principles	cues the mind uses to group parts into whole objects	proximity, similarity, common movement
depth cue	information that helps you judge distance	binocular disparity, interposition
illusion	when what we see does not match reality	Mueller-Lyer, Ponzo, Shepard tables
amnesia	serious loss of memory	the choir director who cannot form new memories
explicit / implicit memory	conscious memory / unconscious memory	last night's dinner / riding a bicycle
semantic / episodic memory	memory for facts / memory for your own life events	"Yale is in New Haven" / your last vacation
recall / recognition	getting a memory out freely / identifying it when you see it	"What color was my tie?" / "Purple or orange?"
Stroop effect	difficulty naming the ink color of a color word	reading gets in the way
change blindness	not noticing big changes in a scene	the man asking for directions is replaced

Check your understanding

1. What are the three kinds of animal communication? Give one example of each.
2. Why does Bloom say "chimps are smart" is not a good reason to expect them to learn language?
3. What two questions are part of the topic "language and thought"?
4. Why is vision a hard problem, for the brain and for computers?
5. Explain why we see the top line as longer in the Ponzo illusion.
6. Which type of memory is each of these: (a) knowing Yale is in New Haven, (b) your last vacation, (c) knowing how to ride a bicycle?
7. What does the Stroop effect show about practiced skills?
8. What do the gorilla video and change blindness studies show about attention?

Answer guide

1. A finite list of calls (vervet monkeys), a continuous signal (the bee dance), and random variation on a theme (birdsong).
2. Smartness is not enough for language. Some smart children, because of a language deficit, cannot speak or understand language.
3. Is language necessary for abstract thought? Does the language you speak affect how you think?
4. The retina is a 2-D surface, but the world is 3-D. Many 3-D scenes fit one 2-D image, so the brain must make unconscious guesses.
5. The converging lines suggest the top line is farther away. Both lines take up the same space on the retina, so the brain decides the farther one must be bigger.
6. (a) semantic memory, (b) episodic memory, (c) implicit memory.
7. A highly practiced skill, like reading, becomes automatic. You cannot stop it even if you want to.
8. Attention is a small window. When we focus on one thing, we can miss big, obvious events without realizing it.

English practice

Useful phrases from the lecture

- "There is absolutely no doubt at all that..." - shows that a fact is certain and not controversial.
- "None of this is to say that..." - prevents a wrong conclusion.
- "What makes it such a hard problem?" - a question that introduces an explanation.
- "And the reason why is that..." - gives a cause.
- "This is one demonstration among many of the fact that..." - links one example to a general point.

Speaking task (with your AI tutor)

- Explain in 4-6 sentences why we see visual illusions. Use: *retina*, *assumption*, *depth cue*.
- Explain in 4-6 sentences what the gorilla video shows. Use: *attention*, *notice*, *change blindness*.

Writing task

- Bloom says, "We are often wrong about our experiences." Write one paragraph (80-120 words) that supports this claim with one example about perception and one about attention.

Study with your tutor

Things to say to your tutor in Live conversation:

- "Let's review Lecture 7. Ask me questions about animal communication, one at a time."
- "Quiz me on the key vocabulary of Lecture 7."
- "I will explain how depth cues create the Mueller-Lyer illusion. Check my understanding and my English."

- "Give me examples of memories, and I will say which type of memory each one is."

Readings from the original course

- Gray, Peter. Psychology (5th edition), chapters 8 and 9
- Hoffman, Donald D. "Visual Intelligence: How We Create What We See." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 81-90
- Hughes, Howard C. "Sensory Exotica: A World Beyond Human Experience." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 91-95
- Schacter, Daniel L. "Searching for Memory: The Brain, the Mind, and the Past." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 165-176

My notes